

element, and no index lookups are required. They are commonly used to store associations in social networks. Graph databases are often faster for associative datasets than traditional databases. They can scale logically to large datasets, as they do not usually require costly join operations. They are a powerful tool for graph-like queries, such as computing the shortest path between two nodes in the graph.

Neo4j

Neo4j is an open source property graph database implemented in Java. It stores data structured in graphs. The graph-based model makes Neo4j highly agile and fast. It is massively scalable, up to several billion nodes, and highly available when distributed across multiple nodes. It can be easily embedded by including the Neo4j library JARs in your build.

Neo4j in high-availability mode has a single master and zero or more slaves—and can handle write requests on all machines, so there is no need to redirect those to the master. A slave can handle writes by synchronising with the master to maintain consistency. All updates propagate from the master to other slaves in due course, so a write from one slave may not be immediately visible on all other slaves.

FlockDB

FlockDB is an open source, fault-tolerant, and distributed graph database licensed under the Apache licence for managing data on the Web scale. Twitter used it to build the user database and manage relationships. It can be efficiently used in high-throughput and low-latency environments. FlockDB was created by Twitter for relationship-related analytics. This database stores graph data that is optimised for very large adjacency lists, and quick reads and writes, but is not optimised for graph traversal operations.

In FlockDB, graphs are stored as sets of edges between nodes, which are identified by 64-bit integers. Each edge between nodes is also marked with a 64-bit position. Edges can be used for sorting. For social graphs, integer node IDs will be user IDs, while in a graph containing favourite tweets, the destination will be a tweet ID.

Key-value stores

Key-value stores store simple key-value pairs, similar to a traditional hash table. The data retrieval paradigm is simple in key/value stores; given a key, they return the value.

Cassandra

Apache Cassandra is an open source distributed DBMS, and an Apache Software Foundation project under the Apache License (version 2.0). It is designed to handle enormous amounts of data spread out across many commodity servers, in traditional or cloud environments, while providing a highly available service with no single point of failure. It

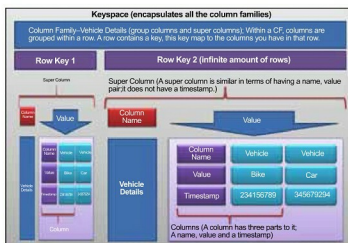


Figure 5: Data model in Cassandra

is a NoSQL solution that was developed by Facebook, and is now used by companies that have large, active datasets such as eBay, Twitter, Reddit, Cisco, OpenX, Digg, etc.

Features include scalability, fault-tolerance, MapReduce support, and a decentralised architecture.

Voldemort

Voldemort is a distributed key-value data store used at LinkedIn for high-scalability storage problems, when simple functional partitioning is not sufficient. It is named after the very popular fictional character from the Harry Potter series—the villain, Lord Voldemort. Voldemort contains in-memory caching with the storage system; hence a separate caching tier is not needed. It supports horizontal scalability for reads and writes. It is more of a fault-tolerant hash table.

Its features are:

- High availability and horizontal scalability for O/R mappers, such as hibernate and active-record.
- Support for distribution across data centres that are far apart, by pluggable data placement strategies.
- Automatic data replication over a large number of servers.
- Versioned data items to maintain and maximise data integrity.
- Transparent failure handling. 

References

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